Can an LLM estimate the embodied carbon of a building enclosure?

Error analysis and prompt design for early-stage A1–A3 estimates on 292 North American buildings

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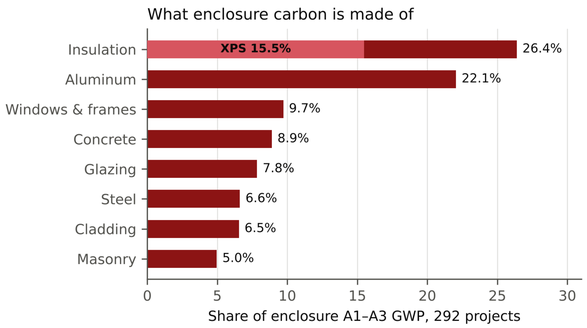
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Question

Exterior-envelope choices lock in much of a building's embodied carbon at schematic design, when no LCA exists yet. Can a large language model, given only early-stage features, estimate the **cradle-to-gate (A1–A3) embodied carbon of the enclosure**, and which prompt design makes it better?

Data & ground truth

- 292 US/Canadian projects, harmonized WBLCA dataset (Benke et al., *Sci. Data* 2025): 928k material rows, 48 early-stage features.
- Ground truth: OmniClass **Shell – Enclosure**, stage **A1–A3**, $kgCO_2e$ per m² floor area. Median 78, range 0.5–514.
- 10 new buildings have no window/glazing data at all: incomplete reference LCAs, flagged.



XPS insulation is 15.5% of all enclosure carbon (43.6 $kgCO_2e$/kg, the HFC-blown product in Tally/One Click LCA of the period).

Method

48 features + glossary → Prompt v1 / v2 / v3 → Claude Opus 5.5 subagents (medium effort) → RANGE low–high → Compare with reference LCA

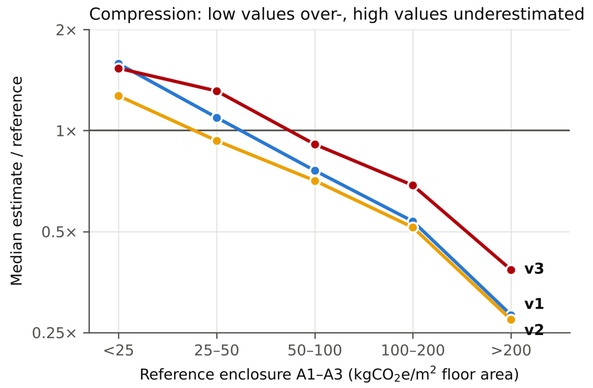
- 3 runs × 292 buildings per prompt; seeded shuffles, identical chunks → paired comparisons.
- Subagents read only their task file: no web, no code, no ground truth.
- Metrics: ±20% coverage, RMSE, |log(est/ref)|, bias (geometric mean), run-to-run CV.
- Held-out test set: 88 buildings (stratified, seed 226).

Baseline: why the model fails

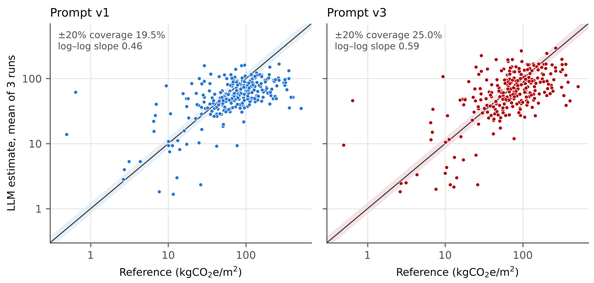
19.5% within ±20% (v1)

0.46 log–log slope (1 = ideal)

0.75 estimate / reference



The model compresses every building toward ~50–60 $kgCO_2e/m^2$: light enclosures are overestimated, heavy ones underestimated 3–4×.



Estimate vs. reference (line 1:1, band ±20%): v3 lifts the cloud toward the diagonal but the top end stays compressed.

Root causes, from 876 reasoning texts

1. **HFC-blown XPS is never considered**: mentioned in 0.6% of answers. XPS explains about half of the bias (est/ref 0.72 → 0.84 when XPS is removed from the reference), and the cold-climate gap too.
2. **Envelope area**: 0.81 when given, 0.65 when the model has to estimate it.
3. **Curtain-wall tail** and **renovations** scoped too narrowly.
4. **The intensity per m² of envelope** drives the error (r = −0.79), and **no feature describes the wall assembly**.

Largest overestimates = incomplete reference LCAs (median est/ref 3.1 for the 10 flagged buildings).

Three prompts

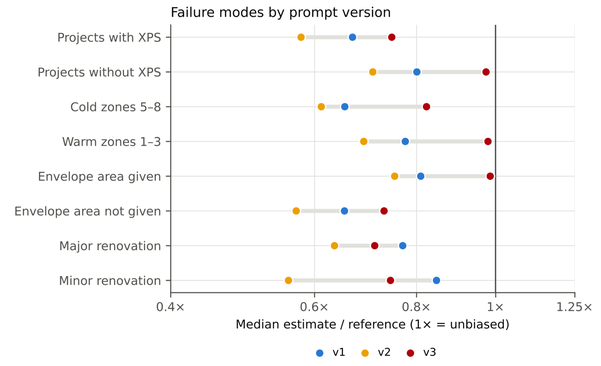
Prompt	Coverage	RMSE	|log| med.	Bias	CV
v1 plain	19.5%	79.7	0.456	0.75	19.0%
v2 method + facts	15.1%	81.0	0.563	0.67	13.8%
v3 anchors	**25.0%**	**76.0**	**0.426**	**0.86**	**12.5%**

v2: scope + step-by-step take-off + named materials

More consistent, but **less accurate**: 89% of answers still assumed polyiso roofs (roof intensity stayed at 35), bottom-up intensities were low (67 vs. 87 per m² of envelope), and "count only new materials" pushed renovations down.

v3: quantitative anchors where numbers are chosen

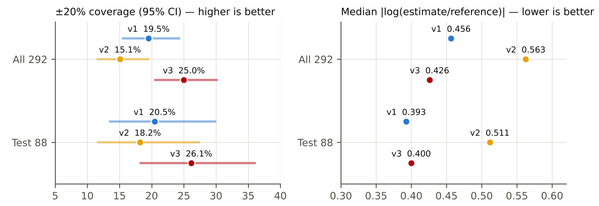
- Component A1–A3 table from EPDs & CLF baselines (walls, glazing, roofs by insulation, aluminum).
- Tool convention: legacy models often use HFC XPS, so include that case.
- Per-envelope sanity check; renovation rule that no longer subtracts.



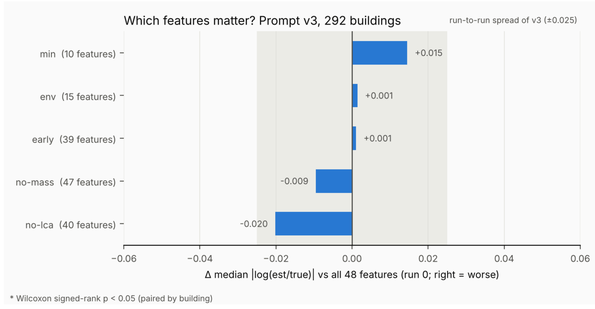
v3 removes the bias for buildings without XPS (0.97), with a given envelope area (0.99) and in warm zones (0.98).

Lesson. Qualitative knowledge in a prompt is repeated but not applied. Sourced numbers at the decision step move the estimate.

Held-out test & features



- All 292: v3 vs. v1 McNemar p = 0.085, Wilcoxon p = 0.053; vs. v2 p = 0.002.
- Test 88: v3 26.1% vs. 20.5% coverage, RMSE 69 vs. 74, not significant (p = 0.41). Every finding behind v2/v3 replicates on the 204 dev buildings.
- **Feature study** (figure below): 10 early-design inputs (climate, use, type, area, storeys, WWR, R-values) are as good as all 48. Removing the leakage-prone `mass_total` does not hurt. All differences are within run-to-run noise.



Takeaways

1. LLMs give usable early-stage enclosure estimates for hotspot screening (¼ within ±20%, bias −14%), but not reliable building-level credit predictions.
2. The error is structured (compression, material blind spots), so error analysis on the model's reasoning shows which prompt fix to make.
3. The ceiling is information, not prompting: adding **assembly type** to the inputs is the next step.

Final prompt (v3, excerpt)

```
1. Areas. Use bldg_therm_env_area if given; otherwise take off roof,
   walls and glazing (bldg_wwr) and state the envelope/floor ratio.
2. Assemblies. Choose wall, glazing, roof; name insulation type and
   thickness. LCA models of this period often model roof and
   below-grade insulation as HFC-blown XPS -> include that case.
3. Intensities (A1-A3 per m2 of component), e.g.
   steel stud + metal panel 70-110 | brick on CMU 135-165
   unitized curtain wall 150-210 | aluminum windows 95-140
   roof + polyiso 20-28 | roof + HFC XPS 85-110 (up to 335-410)
4. Check: below ~50 per m2 of envelope -> something is missing.
5. Range incl. higher-carbon alternatives; do not anchor on a
   typical building.  6. Renovations: do not assume the envelope
   is kept.            RANGE_kgCO2e_per_m2: <low> - <high>
```

Full prompts, code and all 5,000+ model answers: github.com/HeshanChen/CEE226LCA

Data: Benke et al. (2025) Sci. Data 12, doi:10.1038/s41597-025-05216-0 (CC BY 4.0). Intensities: CLF N. American Material Baselines 2023/2025, PCI, Kawneer, RDH EPDs.

Model: Claude Opus 5.5 (medium effort), Claude Code subagents